Welcome to the MacroHive Conversations podcast. I'm Bilal Hafiz, head of strategy at MacroHive. Before we dive in, a quick update. Following MacroHive's acquisition by the BGC Group, BGC has launched a new regulated business called MacroHive Markets. For the first time, onboarded clients can also access execution services via MacroHive Markets. This new workflow will allow macro institutional investors to benefit from strategy and quant insights, alongside access to global liquidity. Please contact Andrew Simon or Gail Mojibriyan Bloomberg or click the link in the description below for more details on McRide markets and its service offering.

Now onto this episode's guest, Stefan Janssen. Stefan is the founder and CEO of Applied AI. He advises Fortune 500 companies, investment firms, startups across industries on data and AI strategy, building data science teams and developing end-to-end machine learning solutions. Before his current venture, he was the partner imaging director at an international investment firm where he built the predictive analytics and investment research practice. He was also a senior executive at a global fintech company with operations in 15 markets, advised central banks in emerging markets and consulted for the World Bank. Now on to our conversation. So greetings and welcome, Stefan. It's fantastic to have you on the podcast show. Thanks for having me. Good to be back. Now, the last time I had you on, we were talking about your book, which was about machine learning and AI for trading. You've essentially rewritten that book, given all the advances that we've seen in AI since over the last three years now. especially agentic AI and so on. So the book kind of incorporates all of that. And we'll talk more about the book later on. But I wanted to kind of use your brain now to see where we are in terms of using AI, which includes generative AI, LLMs, machine learning, kind of this sort of catchall term. Actually, maybe I'll start with that. I mean, how do you define AI? Because it seems to mean lots of different things to people these days. So let's start with that, perhaps. How do you define AI? It's funny, right? There's a saying that AI is always the thing that we don't solve yet. And as soon as we've solved it, it sees as being AI, right? Like maps and the searching of paths in the 80s or so. I mean, it's machine learning is the idea that machines get better at solving a task based on data, you know? So I think that in part is still, you know, related. But of course, large language model is broad and the scope dramatically, right? You know, you have reasoning, et cetera. So ultimately, we always tend to define intelligence as sort of something human inspired, you know, since, you know, for lack of a better model, we ultimately always wanted to define as something machine being as good as humans on a given number of tasks. All of them, some of them are subset, you know. So I think ultimately you converge on something like this really for lack of a better benchmark, you know. Yeah, absolutely. That makes a lot of sense. Now, you know, a big issue kind of I face is that many people use AI for general sort of business purposes, what I would call general business purposes. So, you know, lots of people are, you know, what they call themselves data scientists that they usually work with some kind of data, they run some kind of machine learning, and then they create dashboards, they offer them the revenue and

costs and client churn and that's kind of the bread and butter of a lot of data science. But obviously, financial markets is very different, especially if you use AI for trading, where the bar is way, way higher around the problem trying to solve is way harder. So perhaps you could elaborate a bit on why is machine learning or AI in trading fundamentally much harder than using AI machine learning in kind of general purpose business functions. Yeah, look, there's a lot of pieces to that puzzle. Ultimately, the usefulness or the hardness, let's say, of, you know, applying say machine learning or AI in the context is kind of what is the ultimate use case, right? What do you actually use the model output? If we define it narrowly as kind of machine learning to say predict returns and then trade on these predictions, you know, if you predict your customer churn, you know, then you're probably in a more stable environment, you know, you have competition. you know companies also compete for customers so it's not only in financial markets we have competitive environment margins are very thin information gets incorporated very rapidly so it's quite hard to predict things for longer right because as information you know seeps into the world people act on it and any games disappear instantly so you have that kind of sort of you know perfect competition market kind of situation to the max but then you have a lot of other issues too which is. How many data do you have? What is the effective sample size that you can utilize? And financial markets are not the sort of YouTube environment where you have trillions of cats jumping around. And they all look the same and they don't change. So you have shorter time series. They are not incremented by people uploading data, but they is passing. So one trading session after another adds new data. It doesn't get any faster than that. Maybe by putting more markets online worldwide, But ultimately, there's a limitation to the data that goes in. And so that plus the limited information content in past history as it comes to the future, low noise to signal ratio, makes this particularly hard. So since you ultimately need to prove that you can earn a return on the predictions, and many people are trying that, I think that even though you may be able to achieve some measure of accuracy on your models, And you do, to actually gain from this in the market is hard. So you have this double whammy, right? Hard to predict, even harder to profit from any predictions that you can make. You know, very often you find that you have models, oh, it's really predictive. And then you look at the assets, and then they're very illiquid, you know, they're the microcaps. And then if you were to trade them, you know, then the margin that you sort of predicting they're kind of, you know, it gets eaten up by market impact. You know, so you have a bunch of challenges given that particularly

challenging environments. You know, I suppose one kind of issue you bring up there is one general issue. This is with all quantitative or systematic approaches. There's the risk of overfitting data where the back test looks fantastic, often because you've data fitted, although you don't admit it. And then when you actually run it out of sample in practice, it turns out not to perform as well. Now with machine learning and AI in particular, there's much more of a black box element on top of that as well, which just adds another layer to it where you aren't quite sure why the signal in the back test was generated in the first place. So how do you approach that problem? Yeah, so both being humble and honest go a long way. Humble and always expecting that prediction accuracy is going to be low. So any values that are suspiciously high are usually a major red flag. And we get to all the potential errors that you can build into your historical simulation. But then You have to be honest in the sense that how many things did you actually try to get to the good results that you produced? And that goes back to the point that the effective sample sizes are not huge. So it's much easier to exhaust any signal in there or to just find, you torture the data until it confesses kind of situation, which is harder if you have the billion trillion sample data sets, which we don't have. And so you have to be honest and discount the results. You have all sort of multiple statistics. tools to adjust for multiple testing and the like, which you should certainly use. But once you've done all that, you're still only certain to some probabilistic degree that probably is not chance. Maybe it is. And so you also have to have the ability to forward test anything that you found in the market with very limited risk, as you usually have after you applied all these safeguards and think that after you filtered your tests and think, OK, out of the 100,000 things here, five that actually seem to have a reasonable mechanism. So you can inspect models and we get to tools how you can maybe try to sort of focus more on what mechanism it could be. And then you try to refute it, et cetera. So there are tools for that. After you pass all these different gates, you get to the OK, then let's deploy it and see if it actually works. And maybe you try to start with sort of paper trading and limited capital, et cetera. That, of course, also has trade-offs because most things that you discover

they decay right you cannot afford to do too much of this so you know no free lunch as usual but ultimately it's a mix of trying to detect the noise through various tests both correctness and statistical corrections and then a deployment that shows you that would it actually work in the future where we cannot have any look ahead bias you know and all these other things that you can Build into the back test, which is a historical simulation, right? So the fundamental problem with back test is you try to simulate history. So there are errors in terms of how you build the simulation. Is the data correct? Is it point in time? Was there anything that you used in the past that you wouldn't have known at that time and all these things which. It always sounds trivial when you say it, but when you actually build it yourself, you get surprised by how easy it is to trip over all these things, right? Thinking back in March 2017, would I have known this data point? The label on the timestamp, it says I would have, but is that actually the case? So there are many little things that paint these results. So there's the historical simulation correctness, and then there's the generic part of carrying the future into the… the past into the future and hoping that it stays true in a changing environment, right? So these are the challenges. On the small sample issue, I mean, you said, you know, YouTube camp videos as trillions of them, but, you know, with financial time series, they're just one time series daily. And that's not that much. I mean, have you looked at, I mean, just creating synthetic histories, you know, whether it's through bootstrapping techniques or Monte Carlo or all of those sorts of techniques, do you find that or just synthetic data of some kind through some other process? I mean, do you find that's helpful or does that kind of give you a false sense of security? It is super interesting. We have a whole chapter on this now. We had it before. There was a there was a model timegan that came out of the medical domain actually. It was really well done. It was basically out of the medical domain where you have privacy issues, researchers came up with tools to generate synthetic time series of patient measurements to improve diagnostic tools. And that actually worked in that domain where, oh, patients, humans, we're pretty stable. Our bodies don't change every three months. So you try that in finance, and it's a struggle.

The models get more and more sophisticated, right? Synthetic data are all over, right? Everybody has used nano-banana and image generation and, you know, diffusion, *etc.* So the techniques get much better. But to learn the idiosyncrasies of financial data with, you know, the distribution characteristics, the fat tails, we're actually the outliers matter, right? It's not just about getting 98 percent right. You really want to get the extremes right. That is much harder because the models aren't really built for that. So we haven't seen much and we try a bunch of things in the book to see, you know, could we simulate something? Well, ultimately, again, you know, you want to build synthetic data and see how useful is it in practice, right? Not just run some, some visual similarity tests or just measure some, you know, photosys, et cetera. But actually, if I were to train a model on a synthetic time series, would it be able to predict anything on real data? And these tests are usually very difficult to succeed at. So very interesting. I wouldn't count it out. Things keep getting better. I don't think we're there yet. This is really research frontier. Okay. That's interesting. And then in terms of the explainability side, one common thing people use on the AI side or machine learning side is Shapley values. Do you have a view on that? Because it seems quite elegant. It seems to explain why the output is what it is. But there are some criticisms around it too. So what's your take on that? Does that turn the black box into at least a gray box? Yeah. It's gray to varying degrees. for Parkness. You know, correlation is a problem. You know, Sheppley comes out of game theory, cooperative game theory, assigned sort of the marginal contribution to individual players. Players there are orthogonal or independent data and features are not. So, you know, the beauty of Sheppley values is that they found a very elegant way to compute it. It's a pretty hard problem, but you know, you can set it up computationally so you can get it. And of course, as soon as there's something that's computable that gives you a number, everybody jumps on it. It is very useful and that it shows very clearly how individual features contribute to individual decisions. So I like it a lot, for instance.

Together with the book, we're releasing a few libraries, especially one that focuses on all sorts of diagnostics, like these statistical corrections that I mentioned earlier. And there's also the idea that you build a machine learning model and you predict returns. And then you build a strategy that trades on these returns. And then you want to understand, here's a particularly bad trade. What happened? And you want to see which features actually contributed to that. Were these maybe features, model inputs that usually are not that important, but here they actually cause the problem. Maybe that gives you a hint at how you can diagnose and improve the model, maybe simply by removing that feature or by doing other, you're taking other steps to mitigate that. So I think it's a useful diagnostic tool as long as you're aware that the numbers that it assigned can be off to various degrees of magnitude if features are correlated. So it's not 100% accurate. Noted. But it's a substantial improvement on what was there before. Yeah, yeah. Now, last time we spoke, that was very much the early years of chat GPT. We've moved a long way since then. There's been much more advanced models have been released. There's a whole agentics process now. So in your eyes, how do you use large language models and agentic AI within trading? What are the use cases? How can one use it at whatever stage of the sort of the trading process? It comes in at various points. It does not come in at the point that most people Want them to come in which is take the trading decisions for me make me money. You know, I think that is where it actually does not come in. But if you look at the book we describe this entire thing as usual as kind of workflow or process, you know, you have to come up with some hypothesis, right? There has to be some idea. or a strategy, maybe after you already know in which asset class or market you want to play. Idea generation is something where, you know, LLMs or GNI is, as the name says, it's useful, right? It's not the be all and all, but you know, rather than sitting there for an hour and coming up with stuff yourself, it may get you started, you know, so that's one thing, you know, along the way, you know, it can provide you useful. sort of ideas. If you want to build a model and come up with variations of the model, what else should I try? Which other features could be useful here? Oh, what does astrophysics do? What kind of features does astrophysics actually derive from data? I haven't studied that. Do I want to read the big book? Maybe you give me just 10 ideas. I can then still scan the sources and check if that's plausible. But in terms of pulling in adjacent ideas, I think it's quite useful.

Then on the agentic front, one new chapter in the book is on autonomous agents, of course. And Richwater had this interesting paper last fall. It's their lab, right? The AIA forecaster agent, which they train on prediction markets. And they have an interesting model. It's like a multi-agent system. And you have agents tackle one question. So it's a prediction market. So it's the Fed. rate going up or down at the next meeting and all things like that. What's the CPI going to be like? Is it going to be higher? Usually binary option kind of questions. And so you can set up an agentic system that searches on the web, pulls in information in multiple steps, agents reason about it. Then you have multiple agents that have used, they debate them. You may have a supervisor. In that case, you have a supervisor that sort of looks at all these different opinions, and then if they deviate, then it procures additional evidence to sort this out if possible, and then it comes to a conclusion. And so in this case, they were able to show that it performs actually reasonably well. It doesn't really outperform humans or markets themselves. But what's interesting is that when combined with other information, it generates interesting incremental signal that is actually predictive. So you have machines that are able to process unstructured data, which is really the main value of large language, the very sophisticated and really productive access to a very large amount of text data, where all other methods before were really not very good at. So suddenly you have access to that source, and you can incorporate that into models, into agents, et cetera. So I think that is in terms of direct LLM contribution is the main one. However, there is, of course, the sort of infrastructure or tool contribution, which are coding agents. You know, coding agents, of course, really play a very heavy hand in the process. So the book comes with 50 plus coding agent skills. You know, for agents, you have these skills, you know, that try to define how to correctly implement certain steps. You know, usually the idea is in a company you have a certain workflow and you have your internal process documented and the model I don't really know that because obviously it wasn't trained on internal data. So you create these skills. And then if the model has to execute or the agent has to execute some piece of workflow, it can pull in that information and then apply it. So similarly, you can do this in a quant research workflow. Because you will find that coding agents are excellent at writing software, but they're much less solid when it comes to methodical things on statistics, machine learning, and the like.

many reasons probably simply it wasn't I wouldn't say it because oh it's so much more difficult it's probably that the training data on that domain is much thinner I would assume you know it's especially in finance there isn't that much information shared as in other domains so that could explain some of that But so it helps to kind of beef up the ability to write the actual code with sort of information that tells how to do that correctly. So coding agents in terms of how do I use it? Coding agents themselves are probably the main contributor. But you see that along the process, you can have a bunch of valuable inputs. There's one piece, for instance, knowledge graphs are kind of useful to organize information. And we can talk about those in more detail. Now suddenly you can create these knowledge graphs more cheaply by using LLMs. So there is this kind of second line of defense or second kind of row of contributions of LLMs to just prepare the data, which is an indirect input into the process. So having unstructured data, prepare it, process it so that it becomes more useful in whatever, whether you're a fundamental analyst or a quant researcher, that is helpful. So it's quite versatile. And, you know, leading on from that knowledge graph element that you just mentioned, one thing we've done a lot of work in is using large language model and then adding a rank approach to it, you know, where you kind of link it to your trusted data sets. And then some people have added kind of a knowledge graph on top of the RAG so it becomes like knowledge graph RAG LLM. What's your experience on that kind of area? Because one issue we all face in financial markets is we need to know our sources. Everything kind of has to ideally be audible and reliability and accuracy is super, super important. So RAG approaches have been the path many people have taken. Although there is a debate now whether the context windows are so large you need RAG or not. But it would be great to hear your opinion. Yeah, exactly. Rack is dead, you know, so there was this kind of, I mean, I think the general, so obviously, hallucinations, *etc.* are a major issue. I think we probably already discussed this three years ago. So it's a known issue and it's been, it's been not resolved but addressed and mitigated, you know, by much better, by much improved ability to really

Demonstrate the provenance of claims right so so rec obviously is useful in the sense that you have specific documents that you know make it into the lm's response you know are the agents work right can be just you know one short kind of conversation or an ongoing workflow. So it's useful to infuse that information, either just to keep the model and the responses up to date or to ground it. And then in addition, whatever you get the information from, you ultimately really want to literally pin down the text passage it comes from. Because any decision maker, and that's in finance as well as elsewhere, anybody that really takes an important decision based on LLM input ultimately wants to have the ability to trace it to where does this all come from. And now in practice, you're probably going to do this initially a lot. And at some point when you gain trust in the system, that goes down a little bit. And you're probably not going to check everything. You're still going to spot check. You can also automate that process. But I think the ability technically to do that has increased substantially. And you can, of course, have the LLM reasoning processes have itself verify that these things actually exist. So that altogether mitigates that. The question whether larger context windows make this all obsolete, I'm a little skeptical to be honest. I mean, the haystack finding, et cetera, clearly has improved, but long context windows date, it might be reliability is a little wobbly, you know, beyond the 100, 200K. And many tasks are ultimately require more context than that, you know, which I think in my and that goes back to the use of coding agents and generally this is one of the big weaknesses still of AI generally that many tasks in practice go beyond the, here's a complicated question answer it, you know, it's more like he has a big project that's been working on for 18 months and will continue on who knows maybe forever and it keeps growing and humans work and they build knowledge and memory as they work on it, and LMS don't do that. This is not something you can naturally resolve with just providing more context, right? If the weights of the model don't absorb all this information and start automatically incorporating this. So this is, I think, where RAC and all these things hit the limits, basically, because I think it is the whole quadratic cost of transformers and so that drives all this, right? But

back to the Knowledge Graph, just being able to structure information that is relevant in a useful way that relates different entities to each other, companies and their suppliers, who holds all these securities, *etc.* And do that in a reliable way, of course, allows for other types of queries and for more complex information retrieval ultimately. Having the ability to organize data in that way is great, and having the cost of doing that come down substantially through LLM is, of course, a major win. And, you know, one of the buzzwords of late has been agent harnesses, which in essence is kind of the envelope around the agentic process. I mean, first of all, I mean, in your mind, how do you define a harness? And then what's the right harness or what, you know, in finance, is there specific things we need to keep an eye out on in terms of how we get the agents to interact with each other? Yeah, so the agent harness is sort of literally harness that the agent wears, you know, as it executes its tasks and agents do a ton of things, right? They use tools. They use all sorts of tools. They search the internet. They look through your documents. If you did the open claw thing, it literally is in your bank account. Maybe you better not. But it spins up virtual machines to execute code. So there's a lot of things that the agent can do and The major gains that we've seen in the last and one of the main drivers, I think, of the explosion of coding agents, especially around initially starting with cloud code, but now of course more broadly, is that the efficiency of pool use dramatically improved. which means lower token use, more accuracy, you choose the right tools, the output of the tools is well formatted, the agent knows how to interpret this correctly, take the next steps because they take many steps in this in this execution, right? So errors compound quickly. So the harness is responsible in large part for Increasing the accuracy, so interpreting results, so I'm gonna search the web, right? I get all these results back, how to interpret this correctly, right? So you need instructions for that, similar to what I earlier mentioned with skills, right? Skills that define how to take certain actions and steps correctly. So you can infuse all this in the agent which is driven by an LLM that wasn't necessarily trained on the specific workflow you wanted.

to operate on, like doing research on companies, right? Doing research on obscure companies or on distress debt or real estate or what have you, you know? So the harness kind of guides this. And of course, the coding agent harness is focused on, well, you know, software projects. So you can envision that. maybe i want a better harmless that actually guides us in the specific in a fundamental research process or as a macro researcher right so there probably also things that you can say hmm you know there are certain there's certain logic that i want to imbue maybe internal knowledge right that as a firm you have acquired how to do research properly. So you can kind of embed that as a harness. This is one of the inputs, you know, the other is simply the workflow. What are my tools that I have available? How do I choose among them, et cetera? So very important and more of the gains that we're seeing on that front is driven by post-training and harness improvements than sort of the underlying models getting better themselves probably. And actually that's one of the threads throughout your book actually that and this is not just AI, these AI harnesses is just more generally models that come up with Alpha is one thing, but actually your workflow as a whole from the start of Alpha Generation 2 testing, implementation, post-trade of the trading strategy, that whole workflow is super, super important. And often people don't spend enough time on that. So can you talk more kind of the apps at the higher level around the workflow of building a trading strategy and implementing it? And I think you have something called the ML4T. machine learning, portraying workflow that you talk about in your book. I mean, all of this goes back when I was teaching data science in New York over 10 years ago, a general assembly. And in New York, you always have people from hedge funds, right? There were students from Millennium, Bridgewater, et cetera. I was a partner at investment firm at the time doing things of this nature. And so there's no real domain specific introduction to machine learning. And it really matters because finance is Like other domains right it's domain specific so the data is has its own. Contacts things you have to know about it to interpret them correctly but then also the application of model outputs is very domain specific as elsewhere right if you're in pharmaceuticals you know your models output something you have to know how to do with it so in finance and this is kind of the idea just of this workflow that ultimately you always have to.

at least either think from or incorporate the ultimate application of what the machinery model does, which in trading is okay, we're going to build some sort of trading strategy around this, which maybe we'll predict returns, maybe we'll predict volatility. Now you have models that predict end-to-end portfolio allocations. But as you do all this, you're operating in a market, so you have to take into account all the things that make markets what they are. So there are costs, there's market impact. What can you trade? What are the risks? How do you handle all this? What are tools to mitigate risk? All of that I input into the trading strategy that ultimately influence whether what you do on the machine learning side will be profitable or not. So the workflow is the idea to simply describe all these things, lay out all of these things, see where machine learning or AI can actually help. As I said initially, there are multiple steps beyond just trade and model to predict returns. and then iterate through the process. So the book, what is very new about this book compared to before is that we have nine case studies. So there are like nine case studies that go from the development of a hypothesis or an idea of a trading strategy tied to some data set, be that ETFs or crypto or FX, et cetera, and then go through the entire workflow, right? So, okay, you know, if I'm in this market and this is what the return structure looks like, These are the costs that I will be facing in my situation as a trader. So what is kind of a holding period that I'm looking at? If I have high costs, can I have a five minute holding period? Probably not. Not going to work. So you have things that inform the model design. You're probably not going to predict five minutes returns if you're holding for at least a day. So there are kind of implications of taking into the market context into account of how you design all this. Then, of course, methodological. things about just how you do machine learning, the whole cross-validation process. There are time series data, so you have to apply a certain rigor and certain methods to it. There's the whole statistical machinery of, good, now I have my results. How do I interpret them correctly? What kind of a haircut do I have to apply? Because I've already run 100,000 models instead of just one. And then how do I do this historical simulation of a trading strategy correctly? by taking into account market reality, which if you come from the machine learning world, you may not be super familiar with. Just as if you come from finance, you may not be familiar with all the details of machine learning models. Maybe you've run linear regressions, but now you have more complex models and you have a more industrial approach to this, because machine learning as a sort of tool or discipline has matured. And so part of this process is then also infrastructure.

Since I said many times, if I run a lot of models, I have to be able to account for those. So you have to be able to actually count your trials. So how do you set that up? You're not going to just have a pen and paper on the side and have to drop down a bar for everything you ran. So you have to automate this, store this in a way that's reproducible, countable, so that you can frame the tests correctly, et cetera. And then ultimately, there's the whole question of goods. And now we researched. that's not an end in itself. So how do we actually go to production? So what's also new in the book is that we now talk about how do you actually go live? How do you connect to brokers? So there's new software that does that. And then once you're in production, how do you monitor the strategy? Things change. When do you actually start figuring out that this was not just a bad day, but it actually starts to be evidence that our initial hypothesis is maybe no longer valid. And we should maybe dial it down a little bit. you know, in terms of exposure, *etc.* So you have this kind of end-to-end story literally from idea, research, simulation, ultimately, deployment, production, and then monitoring. You know, hopefully it works forever, but you know, we know how things go. Yeah. Yeah. I mean, one of the challenges I've always found is that when I've hired computer scientists, they don't seem to be as familiar with time series and non-statiary time series. That's been a huge, huge challenge because non-statiary time series brings a whole world of pain when you're doing testing and backtesting. So that's been one of the challenges. Now, in your book, you do talk about the alpha factory, which I assume is linked to this workflow rather than because the risk is you just end up with a bunch of notebooks and you're just kind of knitting them together and kind of hoping that some kind of a robust process. So, you know, is the alpha factory this whole workflow implemented? Is that what you think? Yeah, it is kind of this idea ultimately crystallized in code, you know. So many of these things are driven by configurations. Now, this is one thing where you don't have this one size fits all, you know, so it's not like an alpha factory that you can download. In the book, we illustrate one way of setting this up. Workflows are very idiosyncratic, I always find, which is great for autonomous agents because they can adapt, but you cannot say kind of this is…

This is the ML4T Quant workflow. You got to follow it and here's the recipe. Now there are conceptual stages and there are various ways of implementing them. So for instance, we have like one version of implementing a kind of a model registry and a trial registry for backtests. I mean, there is a bunch of different ways that you could do this and I'm sure that depending on your personal preferences, you will place more emphasis on one aspect on the other and then you create it differently. But the idea is the same. So the logic is that You should start with some idea of what you're trying to accomplish. You don't have to super specifically specify that it's going to be the three-month momentum with a little bit of a volatility kind of flavor around it. You can look more broadly, but you should be prepared to probe these things for an existing mechanism. These are all steps that you would be taking in that. factory kind of style the factory that includes that you know depending on maybe you only run 10 trials if you are really kind of very focused that's rare but more likely going to be thousands of trials you know so there comes the whole filtering process you ultimately the design of your simulation will depend on your personal trading situations you know maybe have very low cost if maybe you're on crypto maybe maybe you're uh somewhere in an exotic emerging market you know all of that shapes what you ultimately build in order to replicate that world historically. But that doesn't change the concepts that you have to obey, or like the boxes that you kind of have to tick. So that is kind of the idea, you know, to make it systematic, to make sure you cover all your bases, and to ultimately to build your own workflow in a way that's reproducible, faster, that you automate what you can automate and use the tools that help you automate, including coding agents, you know. Yeah, yeah, absolutely. That makes sense. Now, in terms of the most of experimental areas as applied to finance, at least, there's some kind of areas that people have been looking at and have struggled with some extent, like reinforcement learnings won. Do you think that's still kind of like an academic thing or does RL work for finance? Yeah, so the second edition, which so six years ago 2020 still had a deep queue network to learn how to trade stocks. So that's gone. So, similar to the idea, generative AI is going to make me money by trading the stock market. Reinforcement learning seems to be made for that, right? You look at AlphaGo, right? If I can win at Go, why can't I win at the stock market, right? But these things don't transfer. The environment is so different, et cetera, et cetera, that this promise of reinforcement learning, I think very few people are pursuing that today.

If you read Sebastian Malaby's book about DeMine's history and Demis Sassabes, and you think with a lot of resources in 2026, could you build something that might get there somehow? Maybe. Would that be something that lasts more than a month? I'm not sure. So is it worth it? I don't know. So I think that idea of reinforcement learning as a trading sort of agent, I think that is hard. However, reinforcement learning, I think even longer than that, has been successfully used for execution. I mean, JP Morgan went live with this in 2017. That is a proper dynamic control problem. So we have three examples in the book instead of the stock trading. One is about execution, which is sort of the mainstay. So how do you slice orders over time? Dynamic control is your x number of shares that you want me to get out how do you place the orders limit markets whatever you know like what are what are the options that you have and how do you ideally sequence them so that works. Market making you know holding inventory etc works somewhat can probably work in practice the smaller examples that we use in the book are not that super successful and then hedging you know dynamic hedging is also quite. successful and accepted as another dynamic control problem where sort of the options positions that you take in order to have, *etc.* are sensible. And so that kind of seems to work, which is so it makes sense conceptually. And I think that is where it has its reinforcement learning really has found a very solid foothold. But the original fantasy of train my little boat and it just runs for me and ding ding ding slot machine, you know, not yet. Another kind of experimental area is world models. You know, Jan Le Koon has been, you know, advocating for world models as the real path to AGI, Fei Fei Li as well, Spatial Intelligence. I mean, they're kind of applying it more to robotics and the physical world, but their kind of idea is that with kind of multi-sensory inputs, not just language and text, and then also working where you kind of abstract states of the world is more efficient and then make predictions about the next stage of the world. I mean, do you think that's a possible pathway for financial market use case there?

i’m very skeptical which goes back to at this stage right i mean who knows what will be possible in 10 years given the speed of development i wouldn’t make any predictions about that robotics is very different in ours like the self-driving cars you can start generating synthetic data about the world you have a very different kind of data situation that allows to train these embeddings. You know, you need substantial amount of data, which in finance, you don’t really have to the same extent. So to me, it seems that the interactive sort of multiplayer world at scale in finance, combined with the relative shortness of history, you know, it’s a lot of data, tick data is large, but you compare it to other domains, it’s limited. I think raise the bar substantially. I think it’s a great idea, you know, world models, it makes a lot of sense, you know, to incorporate additional senses into the perception of agents and it may well be necessary. I mean, it’s a big debate, right? Like, how much content and how much learning can you extract from language alone, you know? So I think it’s a very useful and valid research direction. Obviously, the funding speaks for itself. and their experience. But I don’t think it transfers from robotics very easily to financial markets, a little bit like this reinforcement learning idea. Obviously DeepMind was extremely successful and they went well beyond the Atari games and go for sure. But still, it’s yet another leap into actual financial markets, I think. Yeah. And actually, in terms of the real general intelligence humans, what do you think is the role of humans in the next five to 10 years? I mean, what can we do better than machines, so to speak? So I think we come up with the questions and the design of the goals and evaluations, ultimately, whether something is useful. I think it’s very complementary. And I do think if you work intensely with coding agents, This relationship becomes very, very evident. It is sometimes a struggle enough to find a balance because of certain idiosyncrasies, agents to be helpful, product manager design them. They always want to be helpful, which leads them to do all sorts of mischievous things, which they interpret as being getting us closer to the goal. So monitoring these things, first of all, deciding what to do.

I think it’s something that AI on its own is not necessarily very good at. Once the goal is clearly defined, and for now at least humans, because of all the passive knowledge, the experiential knowledge and all these things that models cannot learn because you never may forget. how much information in the world is not really recorded as of yet accessible to models, right? So how much learning by living really humans absorb in their brain that they can provide as context to a model that models simply for now don't have access to. So that role is absolutely critical. And for now, I don't see models getting there. Now, can you record all these things? but run around all that time with micros on and centers and sure in some futuristic world. I think you can construct that, but I think in reality we're quite a bit away from that. So I think that role is critical, deciding what to do, often how to go about this correctly. If these are processes that are not just written down in some manual, which many things are not, why are certain people experts? Because they have learned something that wasn't learnable before just by reading a book they either discovered it developed itself so they know it and so forth right so i think that role is critical a lot of the execution in between i think models will be able to learn it certain some domains more than others again depending on like with the early example with machine learning and finance it's surprising how many mistakes i still encounter in day to day you know as opposed to building a website, build a website, piece of cake, design, build, deploy it, hands off. So it varies what the execution contribution is of agents as for now, but that's growing very rapidly because the incentives to make all these things accessible grow. So I think that it will be us managing multiple agents in the future with a lot of ADHD developed at the constant. context switching in between these various workflows that we'll have to manage. I think at least if coding and this kind of world is a harbinger of what to come, I think that is where we're heading at least for the near and medium term. I think I'm sure this will further evolve with meta layers of agents managing agents and all that.

Yeah, I mean, one area I've certainly found is that it's kind of current state of AI. It's very good at building demos. But then when you move to production, maintaining the productionized output is, AI really struggles with that. You know, often even with a demo, there's a lot of technical debt or vibe coded. You know, there's bloat, a code bloat. So when you productionize, then agents, I find a struggle a bit. That's where you kind of an experienced hand definitely helps on that side. I think you always, when you drill down into these kind of failures, you almost always find that there's critical context that's missing. And once you start looking at that, you realize just how much context, solving a task that has a lot of diversity, how much context that really requires. which is why people, humans take a long time to develop both as kids, you know, to become adults, but also on the job, right? To become proficient at some activity. It's nothing that you do in a week, right? It often takes years and years and years. And this is that something that we do every day. And it's not like every day is the same. So you accumulate a lot of information. And once you realize, you know, that ultimately that information, which isn't really tangible, You absorb it, but it's real. It exists. You realize how important that is when you see agents failing at these tasks and not incorporating things that you think, yeah, of course, that's kind of obvious. Why don't you apply the convention? Well, because they don't know. So you have a lot of these things still. And you just realize, I think it's marvelous how complex and sophisticated humans are in their ability through lifelong learning. not by reading books but living and doing things, you know, how much knowledge we accumulate and how hard, despite the amazing progress, right? I mean, you look at LLMs and I would just say, I don't know if you've used Fable yet, you know, the latest model. I mean, it's very impressive what it produced, the latest cloud iterated the down sort of the public version of Mutals, right? Very impressive and still. And still it seems to be quite a journey to truly replicate humans. So it's a bit like cognitive dissonance. You have insane progress. Nobody would have expected it. And then you have spectacular faders at the same time. You kind of need to accept both of these as real.

And before we talk about your book, final question, we do have some young people listening to the podcast as well. And many of them will be graduating and joining the job market after the summer. Many of them are quite frightened about job prospects. What advice would you give them about entering the job market in the world of AI, in the face of AI that seems to be eating up lots of junior jobs? Yeah, it's really challenging. And I will not pretend that. I kind of know how to do this, right? I think as a young person, and I was surprised sometimes I teach young kids and do some boot camps, *etc.* I was surprised when I heard of this anxiety, which was like, there is this incentive to do things using LLMs because I'm faster, you know, and the others are doing it. But that means I'm not learning, you know. And I think that conflict is quite real. But then at the same time, you're learning a lot of other things. You're learning how to solve tasks using LLMs. I don't know, and I think it's very domain-specific what the right balance is. I also think that some of these things, they will settle because it will become obvious that at certain tasks, you will not succeed by simply just chat GPT in your way out of everything. I mean, you cannot deliver, I don't know, you want to be an equity analyst, you cannot just chat GPT your way out of this. being young, having the time and the flexibility to use it a lot, you may become a lot better at figuring out where you can use it right and where you cannot use it. And also probably much more open to tracking over time how these things evolve. So I think it's always daunting to be young and step into the drop market. I think anxiety at that point is nothing entirely new. I would look at these advantages and just try to be very very offensive in adopting these tools and not just the chatbot, but definitely get into the coding agent type of world, co-work, and all these things that are more agentic in nature. I think getting on top of that very early and actively figuring out how they can help in your work, but solving real tasks with them, I think it's going to be very, very useful. may also make work more fun down the line because you can just get rid of some of the boring stuff. So there's a lot of positives. That's fantastic. Now, your book, this is, you know, it looks like almost a rewrite of a book you wrote a number of years ago, Machine Learning Portrait. Can you just talk a bit about the book, how you would describe it, and also when it's going to be published and how people could get hold of it? Yeah. So we're less than a month away. Late June, early July will be out on Amazon and packed, et cetera.

So the book is, yes, it's a complete rewrite, obviously sort of same topic, machine learning for trading. The structure is new in that we have these nine case studies. The book itself contains no code anymore. The second edition was still like a bunch of code examples in there. Instead, we have a lot more code outside of the book. So there's like these kind of chapter specific notebooks instead of I don't know. I think the second edition had like 150 or so notebooks. The repo was super popular. I think we're almost at 20,000 stars now. And so we have 150 notebooks in the second. Now we have 250 for the 27 chapters plus another, I don't know, almost 200. covering the case studies. And so there's much more emphasis on covering the steps of, you know, this workflow that I described a few times, much more systematically. The second edition was like, well, here's gradient boosting. Okay. So, you know, here's a data set, ETFs, let's just build a model that uses gradient boosting to break returns and then let's back test something. So it was very, very simple, you know, now you have all of this systematically. All the way through what if i apply risk management what if i apply cost sensitivity analysis what what if i vary the model right instead of linear regression i use gradient boosting i use deep learning i use latent factor models what happens if i apply cause inference you know what does it tell me about the validity of the features that i'm using and the stability of the relationships. You know, so it goes into much more depth, much more systematically compares this across strategies, which are different asset classes. So we cover US equities, both intro day and longer holding periods. There's crypto, of course. There's FX. There's futures ETFs. I think that's about it. And so nine case studies, five or six asset classes. six libraries that come with the book that is of course a test to monitor the productivity of coding agents that cover everything from data sources to kind of feature engineering and testing to statistical diagnostics of machine learning models and strategies. There's a models library that embeds or that allows you to much more easily apply models that are only published in research, you know, like more recent late inspector models or

some end-to-end portfolio learning models, and then there's a new backtesting library. I used to maintain this zipline backtesting library, that Quantopian. Quantopian was a browser of hedge fund. They unfortunately closed down three months after I published it back in the edition. and it used their tools, so I maintained that for a while, but that was very old. They have that, and then you have a library that takes everything live via interactive brokers or I'll pack on it. So that's kind of an end-to-end workflow there with supporting tools. So that is new. There will be new educational offerings. So there are courses on Maven. One is, and so we cover a gen AI tool, right? So there's stuff about retrieval of mental generation, knowledge graphs, autonomous agent, the forecasting agent. We have an agent sort of that tries to implement quantum research. So the idea is that we have this structured workflow with well-defined artifacts that come out of this, like model results, back test results, et cetera, that agents can interpret because they're standardized. Then they have access to standardized tools, which are these libraries that execute all these things much more deterministically. So subject to the vagaries, oh, is the agent going to run the test correctly? It just calls the tools that has already. coded, right? So you learn how to build something like that, which allows you to iterate on these strategies because obviously the book only interprets or the book only sort of illustrates a basic implementation, you know, like, well, you know, if you get first started to take these features and these labels and these models, *etc.* But there's a lot more, of course, that you can do to make it profitable. So there's all that. And so the educational offerings are kind of, there's going to be a workshop on how to build such agents. And there's a much more in-depth hands-on course on how to take all these case studies, develop them further, or develop your own trading strategies that's going to be on Maven. So it's a much bigger package because obviously content generation has become easier these days. And I was in part initially I was wondering when I embarked on this six months or so ago, today in 2026, why would you still write a book and not just say provide a curated set of prompts? Because the specifics are readily available. But then, of course, you still have

you know, like what is the role of humans? You know, you have the benefit of curation. And once you try to do that with LLMs, you realize how difficult it is for them to operate at that scale. The book has 750 pages, you know, and to kind of tie these different pieces together coherently is a major challenge for LLMs, which I would not have expected to be honest. So that was actually quite an interesting experiment. So that it's new. It uses the technology but with substantial human touch in between. And as I said, it's agent-friendly. The libraries are agent-friendly. So they have all the coding agent documentation in them. And it comes with these coding skills. So it tries to be a lot of things in 2026. That's fantastic. I mean, I've seen an advanced copy of the book. It's really, really good. The libraries are excellent as well. So I'd urge everybody to get hold of the book. I'm going to… give it to everyone in my team as well so that we kind of implement best practices. So congratulations once again on writing such an excellent book. And I think it's one of the only ones as far as I can tell in this space that's so practical. So well done on that. And thanks again for coming on this podcast and good luck with all your endeavors. Thank you so much for having me. All the best to you. Thanks for listening to this episode. Please subscribe to the podcast show on Apple Spotify. You'll be able to listen to podcasts, leave a five star rating, a nice comment and let other people know about the show. We'll be very, very grateful. Sign up for free newsletter at macrohype.com. Macrive Ltd. Macrive retains all ownership title rights and